Scholarship of Teaching and Learning

Teaching Innovations, Learning Resources, and Publications

Prof. María del Mar de Fez Laso

Teaching and Learning Publications

This portfolio brings together teaching innovations, mathematics practice resources, and educational outreach publications developed by Prof. María del Mar de Fez Laso. Across these areas, her work emphasizes active participation, independent practice, reflection, reasoning, and communication.

Teaching Methods

- Educational Innovation Series - Volume 1

  An overview of teaching approaches developed to encourage active participation, independent learning, and scientific reasoning. It brings together practical ideas for supporting engagement and making learning progress visible.

  https://doi.org/10.5281/zenodo.21311491

- Educational Innovation Series - The IRC Framework

  Introduces the Inquiry–Reflection–Communication Framework, connecting investigation, reflection, and communication to develop transferable skills.

  https://doi.org/10.5281/zenodo.21338475

- Educational Innovation Series - Stand-Up Teaching

  A collaborative teaching approach in which students work together on problems in a safe, supportive environment. It encourages discussion, shared reasoning, and the development of self-confidence, while enabling instructors to identify difficulties on the spot and provide immediate guidance.

  https://doi.org/10.5281/zenodo.21377654

- Educational Innovation Series - Blue Points

  A system for recognizing students' contributions to classroom learning. It encourages participation, explanation, and communication while valuing repeated small contributions.

  https://doi.org/10.5281/zenodo.21383654

- Educational Innovation Series - Learning Checkpoints

  Brief opportunities for students and instructors to assess understanding during learning. They help identify gaps and guide the next steps in practice and teaching.

  https://doi.org/10.5281/zenodo.21384248

- Educational Innovation Series - A Feedforward Approach

  An approach that uses feedback to guide students' next attempt. It emphasizes actionable suggestions that learners can apply to improve subsequent work.

  https://doi.org/10.5281/zenodo.21393587

- Educational Innovation Series - From Calculation to Reasoning

  An approach that extends mathematical tasks beyond obtaining an answer. It encourages students to explain their methods, justify decisions, and communicate their reasoning.

  https://doi.org/10.5281/zenodo.21396109

Teaching Methods (continued)

- The Inquiry–Reflection–Communication (IRC) Framework: A Transferable Model for Developing Future-Ready Skills in Higher Education (Expanded AUBH Research Day 2026 Presentation)

  An expanded presentation of the Inquiry–Reflection–Communication Framework and its application in higher education. It connects the framework's stages with opportunities to develop transferable skills.

  https://doi.org/10.5281/zenodo.21497169

- Educational Innovation Series - The IRC Framework (Mandarin version)

  A Mandarin-language infographic introducing the Inquiry–Reflection–Communication (IRC) Framework and its applications across disciplines. It makes the framework, originally published in English, accessible to Chinese-speaking educators, extending its reach across linguistic and cultural boundaries.

  https://doi.org/10.5281/zenodo.22995244

Mathematics Exercise Banks

- College Algebra, Real Numbers, Exercise Bank with Solutions

  A structured collection of 484 exercises covering 44 distinct exercise types, each with ten alternate versions. Answers and selected worked solutions support independent practice and self-assessment.

  https://doi.org/10.5281/zenodo.23182292

- College Algebra, Variable Expressions, Exercise Bank with Solutions

  A structured collection of 330 exercises covering 30 distinct exercise types, each with ten alternate versions. It develops skills in evaluating, simplifying, translating, and applying variable expressions.

  https://doi.org/10.5281/zenodo.23138180

Women in Science and Educational Outreach

- Women Who Shaped Astronomy

  An educational outreach publication highlighting women's contributions to astronomy, developed as part of the validation of the IRC Framework. It broadens the visibility of scientific role models and encourages engagement with the history of discovery.

  https://doi.org/10.5281/zenodo.20398087

- The Women Have Got the Formula - Dari & Pashto Edition

  A multilingual outreach publication highlighting women's contributions to chemistry. Based on the original publication *Ellas tienen la fórmula* by Ciutat de les Arts i les Ciències (Valencia, Spain), this edition presents selected content in English, Dari, and Pashto through a collaborative translation and dissemination initiative aimed at supporting scientific awareness and educational accessibility.

  https://doi.org/10.5281/zenodo.20412335

About the Author

Prof. María del Mar de Fez Laso's work in the scholarship of teaching and learning focuses on helping students participate actively, practice independently, and develop confidence in reasoning and communication. She encourages them to question, explore, and learn through trial and reflection. Her teaching is grounded in attentive listening, hands-on learning, and genuine care for each learner.

She is the creator of the Inquiry–Reflection–Communication (IRC) Framework and has developed teaching approaches including Stand-Up Teaching, Blue Points, Learning Checkpoints, A Feedforward Approach, and From Calculation to Reasoning. Her publications also include College Algebra exercise banks and educational outreach works such as *The Women Have Got the Formula* and *Women Who Shaped Astronomy*, which promote inclusion, representation, and engagement with science.

For more than twenty years, Prof. de Fez Laso has taught mathematics, physics and astronomy. She has held senior academic and leadership positions in Bahrain and Afghanistan, including Distinguished Professor, Dean of Research and Graduate Studies, Chair of the Division of Science, Technology and Mathematics, Program Lead, and Director of Scientific Outreach. These international experiences have strengthened her belief that effective education must combine academic rigor with accessibility, practical engagement, cultural awareness, care and respect for every learner. This commitment also motivates her educational outreach and other work in several languages, making scientific ideas and teaching approaches accessible to audiences across linguistic and cultural boundaries.

Prof. de Fez Laso received her degree and PhD in Physics from the University of Valencia, Spain, specializing in experimental particle physics. She conducted her doctoral and postdoctoral research at CERN, contributing to major international experiments, and has published more than 90 scientific articles in particle physics. Her career has also included nine years in investment banking, giving her a valuable perspective on the application of quantitative reasoning in a real industry setting and beyond academia. Beyond education, her great passion is traveling, through which she continues to learn from different people, cultures and ways of seeing the world.

Author Information and Licensing

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